TECHNICAL EVALUATION REPORT

Memory Rank and Liftability of Transition Families Without a Generator: A Projection Theory for the Indivisibility Deficit

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1 Summary of the submission

The paper studies two-time families $V = \{V(j \leftarrow i)\}_{i<j}$ of column-stochastic kernels on a finite state space over a finite time grid, and asks how to convert the failure of the Chapman–Kolmogorov (CK) property into a well-defined, stable, computable invariant.

The argument proceeds in three steps. Section §2 reads CK as flatness of a transport over the complex of time intervals, treats the defect $\Delta_V(t,s,r) = V(t \leftarrow s) - V(t \leftarrow r)V(r \leftarrow s)$ as a kernel-valued 2-cochain, and derives the coherence identity that associativity forces on it. Section §3 closes the obvious route: re-dressings $V(j \leftarrow i) \mapsto g_j V(j \leftarrow i) g_i^{-1}$ act on the defect by conjugation and cannot annihilate it, but the closure of every orbit contains flat families, so no orbital minimisation of a flatness-distance defines an invariant. Sections §4–§5 build the invariant instead by metric projection: within a declared dilation category, F_d is the *closure* of the set of observable compressions of flat transports on d hidden states, the liftability profile is $h_d(V) = \operatorname{dist}(V, F_d)$, and the memory rank is the least d at which it vanishes. Section §5 adds a realization theorem with an explicit dimension bound and a computable obstruction separating families that are the conditional family of no process. Section §6 develops first-order theory at the projection, §7 shows the profile is category-relative, §8 reports finite-state computations, and §9 states open problems.

The substantive standards in force are logical consistency with a testable hypothesis, and reproducibility for the empirical part. I have judged the paper on three axes: whether the theorems are correct and correctly scoped; whether §8 establishes what it is said to establish; and whether the construction is new against the realization-theory and metric-algebraic-geometry literature it places itself in. Every quantitative statement below rests on an independent computations, performed from the printed definitions alone.

2 Overall assessment

The paper features a highly structured framework driven by its negative results. Two central negative findings serve as foundational components for the overall argument. Theorem 3.3 rules out an alternative gauge-based approach that a reader might otherwise assume viable. Theorem 5.6 establishes a lower bound on the construction by demonstrating families that cannot be explained by a dilation of any dimension. Elevating these limitations to formal theorems is an unconventional but effective approach; it systematically eliminates alternative methodologies, thereby justifying the validity and necessity of the proposed positive construction.

The theorems exposed are correct as a mathematical statement. Several formulas can be easily checked: the coherence identity, the inverse $g_i^{-1} = \varepsilon_i^{-1}(I - (1-\varepsilon_i)w\mathbf{1}^\top)$ and the conjugation formula of Theorem 3.3 together with its entrywise admissibility bound. Beside that the splitting construction behind the nesting proposition, the Lipschitz constant of the consistency defect, and the history lift of Theorem 5.7 seems to be correct and I couldn't find errors on that respects.

What the theorems are said to mean, however, is in three places stronger than what they say, and in one place the numerical section is said to establish something it cannot.

First, the model sets are closures, so a vanishing profile certifies approximate liftability and not the existence of a lift (Finding 2). F_d is defined as $\operatorname{cl}\kappa(\Theta_d^+)$, and Θ_d^+ requires strictly positive observed block masses. The abstract nevertheless states that the zero stratum "consists exactly of the families that extend to a flat transport on a d-state structured enlargement". That equivalence is not proved and is false as stated: One can exhibit a family with $h_2 = 0$ that no lift in Θ_2^+ realizes. The gap is operational rather than formaland the closure is forced, since attainment in

Proposition 4.3 requires a closed model set. What must change is the reading, not the definition.

Second, the numerical section cannot certify the ranks it reports (Finding 5). Section §8 states correctly that its values are certified *upper* bounds on h_d. An upper bound $h_3 \leq 0.026680$ does not establish $h_3 > 0$, so the claim that the memory rank is "recovered" at four is not certified by the computation that is offered for it. Multistart dispersion and first-order stationarity are evidence of a good local minimum, not a global non-membership certificate, and the model set is nonconvex and semialgebraic. This is complementary to a second failure that seems to exist and can be measured independently: above the memory rank the reported profile violates the paper's own monotonicity proposition (Finding 6). The numerical determination of the rank is therefore uncertified on both sides (the positive values below it are unproven and the zeros above it are unreliable).

Third, the claim made for the computations is false, and the paper's own Proposition 4.4 refutes it (Finding 4). The memory rank is the *minimal* realizing dimension, which is what Definition 5.1 says and what the invariant should be; it is not the generating dimension, and the splitting construction used to prove Proposition 4.4 produces the counterexample in one step.

Fourth, the consistency defect gives a one-sided obstruction only, and sectino §8.2 uses it as though it were two-sided (Finding 3). $\kappa_c(V) > 0$ certifies non-realizability; $\kappa_c(V) = 0$ does not certify the converse, and §8.2 invokes Theorem 5.6 for a family merely asserted to be process-free without reporting κ_c at all.

Both necessary corrections have been provided. First, a computation yields $\kappa_c = 0.324012$ for a random family, which confirms the existence of the missing certificate. Second, a specific family is presented where $\kappa_c = 0$ exactly, demonstrating that it cannot be the conditional family of any process. Along with this example, a structurally stronger obstruction is introduced; this obstruction is computable via linear programming and yields the same Lipschitz argument and conclusion.

To these must be added the reproducibility of §8, which the data statement asserts and the text does not deliver (Finding 7); two local errors, in the boundary statement of §6 and in the quantitative bound of Proposition 7.1; and a group of positioning items (Finding 8).

3 Section-by-section analysis

3.1 The deficit complex (§2)

The metric (4) is well posed and the flatness definition is correct. The observation that $\mathbf{1}^\top \Delta_V = 0$ is correct and is used consistently.

The coherence identity is correct, and the proof is the two lines claimed. Writing out $\Delta_V(t,s,r) - \Delta_V(t,s,q) = V(t \leftarrow q)V(q \leftarrow s) - V(t \leftarrow r)V(r \leftarrow s)$ and substituting the definitions of $\Delta_V(q,s,r)$ and $\Delta_V(t,r,q)$ cancels the two triple products and leaves (5) exactly. Every deficit appearing in the identity is indexed by an admissible ordered triple. The manuscript's own use of the identity as a correctness check on any implementation that estimates deficits from data is good practice that was useful for checks.

The higher coherence is asserted, not proved. Remark 2.3's reading of (5) as $\delta_V \Delta_V = 0$ for the displayed twisted coboundary is correct, and the identification of the four-point instance with the pentagon of K_4 is a reasonable analogy. The sentence that follows: that "the higher identities on five and more points follow from (5) by the same substitution scheme" asserts a coherence tower

that is nowhere defined: no complex of higher cochains is given, no higher coboundary is written down, and $\delta_V^2 = 0$ is neither stated nor proved. Either define the complex and prove the identities, or keep the associahedron reading explicitly as an analogy. **[R]**

3.2 The gauge route (§3)

Finding 1 (The trivialization theorem is correct; its reading and its group both need work)**.** Theorem 3.3 is correct in every detail I checked. With $\varepsilon_i = \varepsilon^{i+1}$ and $g_i = \varepsilon_i I + (1-\varepsilon_i) w\mathbf{1}^\top$: $\mathbf{1}^\top g_i = \mathbf{1}^\top$ because $\mathbf{1}^\top w = 1$; the stated inverse is verified by direct multiplication using $(w\mathbf{1}^\top)^2 = w\mathbf{1}^\top$; the conjugation reduces, using $\mathbf{1}^\top A = \mathbf{1}^\top$, to $\rho_{ij} A - \rho_{ij}(1-\varepsilon_i) Aw\mathbf{1}^\top + (1-\varepsilon_j) w\mathbf{1}^\top$ as displayed; and the entrywise lower bound $(1-\varepsilon)\min_b w_b - \varepsilon$ is valid because Aw is a probability vector. Corollary 3.4 follows because a continuous function constant on an orbit is constant on its closure. The result is valuable precisely because it is negative.

Three things belong with it.

i) The invariant-theoretic reading exceeds the theorem. Remark 3.5 asserts that "every point of $\mathcal{V}$ is unstable for this action", that "the categorical quotient collapses", and that "the ring of continuous invariants vanishing on Flat is trivial". Instability, categorical quotients and invariant rings are statements about an algebraic group acting on an affine variety, and none of the three objects is specified here: the admissible set depends on positivity and varies with the point, the acting set is not a group action on a linear representation space everywhere, and a ring of *continuous* invariants is not a geometric-invariant-theory object at all. What is proved is Corollary 3.4, and it is enough for the paper's purpose. The language should be narrowed to it, or the algebraic framework supplied.

ii) The arithmetic floor should be recorded. Below $\varepsilon \approx 3 \times 10^{-3}$ on a five-point grid the computed defect stops decreasing and rises again because g_i^{-1} amplifies rounding error by $\varepsilon^{-(m+1)}$. The trend is an artefact of arithmetic, not of the theorem, but a reader reproducing §8.4 will meet it.

iii) The group has a name. The re-dressings of Definition 3.1 (invertible g_i with $\mathbf{1}^\top g_i = \mathbf{1}^\top$, acting by conjugation across times) are exactly the change-of-basis transformations relating two quasi-realizations of the same process in hidden-Markov realization theory, the framework of the surveys the paper cites for realization. Making the identification would strengthen the theorem rather than weaken it: it converts a construction that looks ad hoc into a statement about a recognised object, and it raises the natural question of how the trivialization interacts with the known non-uniqueness results there. **[R]**

3.3 Model sets and the profile (§4)

Proposition 4.3 is correct as well as its proof. Proposition 4.6 is exact.

The nesting proposition is correct, and I verified its construction. The splitting of a hidden state into two copies with identical source columns satisfies the Kemeny–Snell strong lumpability criterion, so the merged chain is Markov with the original matrices and initial law, and the observed process is unchanged.

Proposition 4.5 is correct: Θ_d^+ is semialgebraic, the compression is rational with positive denominator on it, the image is semialgebraic by Tarski–Seidenberg, closures and partial minimisations of semialgebraic objects are semialgebraic, and Whitney stratification is classical. The phrase "it is 1-Lipschitz squared-root by Proposition 4.6" in the proof is garbled and should be repaired. **[R]**

Finding 2 (A vanishing profile certifies approximate liftability, not liftability)**.** Definition 4.2 sets $F_d = \mathrm{cl}\,\kappa(\Theta_d^+)$, so

$$h_d(V) = 0 \iff V \in \mathrm{cl}\ \kappa(\Theta_d^+),$$

which says that V is approximated arbitrarily well by exact d-state compressions with strictly positive observed block masses. It does not say that some $\theta \in \Theta_d^+$ has $\kappa(\theta) = V$. The manuscript reads it the second way throughout: the abstract says the zero stratum "consists exactly of the families that extend to a flat transport on a d-state structured enlargement"; the significance statement says the profile "vanishes exactly at the minimal hidden dimension"; Definition 5.1 is glossed as the minimal hidden dimension. The equivalence is not proved, and it is false as stated.

A witness, at the smallest possible dimension. Take $n = 2$ and the one-step kernel $M = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, which is column-stochastic with $M^2 = M$. The family $V(j \leftarrow i) = M$ for all $i < j$ is flat, and by Example 5.2 $F_2^{\mathrm{inhom}} = \mathrm{Flat}$, so $h_2(V) = 0$. An exact lift in Θ_2^+ would require $p_i(\theta) > 0$ at every time; but $M\pi = (1, 0)^\top$ for *every* initial law π, so the second block mass vanishes at time 1 and no such lift exists. Hence $V \in F_2 \setminus \kappa(\Theta_2^+)$, and $h_2(V) = 0$ certifies closure-liftability alone.

The gap is operational, not formal. The compression formula divides by block masses, so sequences may approach boundary points at which a block mass vanishes while the conditional columns survive as limits. The manuscript itself reports projections contacting the boundary with block masses down to 10^{-13}. I measure the same effect independently: for a family assembled from independent random kernels, the attaining lift at $d = 3$ has smallest block mass 6.9×10^{-9}, while for a genuinely realizable family the attaining lifts are interior (smallest block masses 5.6×10^{-2} at $d = 2$ and 3.0×10^{-1} at $d = 3$). Whatever $h_d = 0$ means, it is being approached through the boundary in exactly the cases the paper cares about.

The closure is not a mistake. It is forced: Proposition 4.3 needs a closed model set for the distance to be attained, and without attainment the profile is not the object the rest of the paper uses. So the repair is not to the definition. Two clean repairs are available. The terminological one: call the invariant a closure-liftability or approximate-liftability rank, and say plainly that $h_d = 0$ means approximability at hidden dimension d. The structural one: enlarge the lift category to admit boundary lifts with vanishing block masses, specify how conditional columns on a zero-mass block are represented, and prove that the resulting exact image is F_d; the manuscript's own Lemma 5.4, which passes marginal consistency to limits, is the template for such an argument.

And a question the paper should answer either way. My witness has a degenerate one-time marginal. Is the gap between $\kappa(\Theta_d^+)$ and its closure confined to families whose marginals degenerate, so that the manuscript's reading is correct on the families it cares about? That is a precise questionand settling it would either restore the abstract's sentence or delimit it. **[M]**

Stability is not a rank test. Proposition 4.6 and Lemma 4.8 are correct and correctly argued, and Remark 4.7 draws the right consequence *for the profile*: if $\mathrm{dist}(\widehat{V}_N, V) \to 0$ then $h_d(\widehat{V}_N) \to h_d(V)$ in the same mode and at the same rate. It does not follow, and the manuscript does not claim but a reader will infer, that the memory rank is consistently estimable. The rank is an integer read off at the discontinuity $h_d = 0$; a consistent decision rule needs a threshold sequence and a positive margin separating the true family from F_{d-1}, neither of which is supplied. Estimation noise moves a family off a model set it lies on, so the plug-in rank is biased upward. One sentence distinguishing stability of the distance from consistency of the rank belongs with Remark 4.7. **[R]**

3.4 Memory rank and realization (§5)

Example 5.2, verified. The identification $F_n^{\mathrm{inhom}} = \mathrm{Flat}$, and hence $h_n(V) = \mathrm{dist}(V, \mathrm{Flat})$, is correct: for $d = n$ every labeling is a bijection and the compression is the lifted family relabeled.

One step of the argument is missing and should be supplied (it is also the step that produces the witness of Finding 2). Membership in Θ_d^+ requires strictly positive observed block masses, so a flat family one of whose kernels has a zero row is the compression of no lift in Θ_n^+. The identity survives because F_n is a closure and flat families with strictly positive marginals are dense in Flat, but as written the proof does not address the positivity constraint. **[R]**

Proposition 5.3, the Łojasiewicz comparison. Correct in both directions. The lower bound follows from Lipschitz continuity of the polynomial map $V \mapsto \Delta_V$ on the compact family space evaluated at a nearest flat family, and the upper bound is the Łojasiewicz inequality for the semialgebraic function $\|\Delta_V\|^2$ with zero set Flat. Combined with Example 5.2 this calibrates the bottom of the profile against the deficit norms, as claimed. Both constants and the exponent are non-constructive, so the comparison is qualitative; for the applications the programme envisages, a bound on θ in terms of n and the grid would be worth more than the statement.

Finding 3 (The consistency defect is one-sided, and a stronger computable obstruction is available)**.** Lemma 5.4 and Theorem 5.6 are correct so far I could check. Every exact compression has compatible one-time marginals and the property passes to limits; the consistency defect is Lipschitz in the family with the stated constant (the displayed $C_* = n(\min_{i<j,a} p_{ij}\pi_i(a))^{-1/2}$ is valid, if slightly loose, since $\sqrt{n}$ suffices) so evaluating at a nearest point gives $h_d(V) \geq \kappa_c(V)/C_*$ uniformly in d. This is the paper's most directly usable result: it is computable from the data alone by a small optimisation, it needs no projection, and a positive value certifies in advance that no dilation of any dimension will explain the family.

The implication runs one way only, and §8.2 uses it as though it ran both. The section asserts that a family assembled from ten independent random kernels "is the conditional family of no process; Theorem 5.6 predicts a profile bounded away from zero". Theorem 5.6 predicts nothing of the kind from non-realizability: its hypothesis is $\kappa_c > 0$, and κ_c is never computed. The inference as printed is invalid.

The certificate exists: For a family assembled the same way I could compute $\kappa_c = 0.324012$, hence a certified floor $\kappa_c/C_* = 0.162$, against a measured profile $h_2 = 1.139742$, $h_3 = 1.035487$, $h_4 = 0.809434$. The same computation on a process-induced family returns $\kappa_c = 1.9 \times 10^{-5}$, numerically zero, as Lemma 5.4 requires. So the missing line is one line, and §8.2 should print it.

But $\kappa_c = 0$ is not sufficient, and the gap is not exotic. Marginal consistency is a pairwise condition; it does not assemble a joint law. One can built families with $\kappa_c = 0$ *exactly* (consistent marginals imposed by construction, residuals at 5.6×10^{-17}) on $n = 2$ and the grid $\{0, 1, 2\}$, and tested realizability by asking whether any probability distribution on $\{1, 2\}^3$ reproduces all three two-time joints. For the family displayed the linear program is infeasible, with minimax compatibility defect $\delta(V) = 0.003332$: consistent marginals, and no process. Of forty independently built marginally-consistent families, nine were non-realizable in the same sense.

And the obstruction strengthens without effort: Define

$$\delta(V) \;=\; \min_{p \in \Delta([n]^{m+1})} \; \max_{i<j} \; \max_{a,b} \; \big|\, \mathbb{P}_p(X_i{=}a, X_j{=}b) - V(j \leftarrow i)_{ba}\, \mathbb{P}_p(X_i{=}a) \,\big|,$$

a linear program in p. It vanishes on every exact compression, because the observed process supplies the joint law; it therefore vanishes on F_d by continuity, exactly as κ_c does in Lemma 5.4;

and it is 1-Lipschitz in V because the marginals are bounded by one. The proof of Theorem 5.6 then gives,

$$h_d(V) \;\geq\; \delta(V)/C' \qquad \text{for every } d \geq n,$$

with $\delta \geq$ a constant multiple of κ_c and strictly greater on the families above. This is a strict strengthening of Theorem 5.6 at the cost of replacing one small optimisation by another, and it closes the sufficiency gap in the direction that matters for §8.2: a positive δ certifies the plateau where a zero κ_c certifies nothing. **[M]**

Theorem 5.7, the realization theorem. The history lift is correct. The hidden space is the disjoint union of the layers of positive-probability histories, the chain occupies layer k at time t_k so the states outside the current layer carry zero mass and the arbitrary assignments are immaterial, the columns are stochastic, and the labeling sending a history to its last symbol makes φ of the hidden state at t_i equal to X_{t_i}, so the block conditionals are exactly $V(j \leftarrow i)$. The bound $D = \sum_k N_k \leq \sum_k n^{k+1}$ is correct and, as the manuscript says, crude.

The final clause is wrong, and has a clean repair that seem to be better than deletion. "On this set the limit is attained at the finite dimension D" is asserted of the closure of the process-induced families, where D (defined from the positive-probability histories of the inducing process) need not exist, since a boundary point of the closure need not be process-induced. The repair is to use the *universal* bound: every process-induced family on an n-state grid with m steps has $D \leq D_{\max} = \sum_{k=0}^{m} n^{k+1}$, so all of them lie in $F_{D_{\max}}$, and $F_{D_{\max}}$ is closed; hence $h_{D_{\max}}$ vanishes on the whole closure. The statement the paper wants is therefore true, with $D_{\max}$ in place of the family-specific D. **[R]**

Finding 4 (The profile recovers the minimal realizing dimension, not the generating dimension)**.** The abstract states that the computations "recover the hidden dimension of lumped Markov chains exactly", and section §8.1 that the profile "does not merely decrease; it vanishes exactly at the generating dimension". Neither holds in general, and the counterexample is produced by the paper's own Proposition 4.4.

Take any three-state chain and split one hidden state into two copies by the construction used to prove that proposition: two copies with identical source columns, the incoming mass divided equally. The result is a four-state chain. Lump it two-and-two so that the two copies lie in one block. By the proposition the observed family is unchanged (I verified this, obtaining agreement to 2.2×10^{-16} between the family observed from the four-state chain and the family observed from the three-state chain) and $h_3 = 0$ for it, witnessed exactly by the three-state chain itself. Generating dimension four, memory rank three.

Nothing is wrong with the invariant. Definition 5.1 defines the memory rank as the *least* dimension at which the profile vanishes, and minimality is the property one wants: an invariant returning the generating dimension would not be an invariant of the observed family at all, since the same family is generated by chains of every dimension above the minimum. What is wrong is the claim made for it. The correct statement is that the memory rank is bounded above by the generating dimension and equals it for generic lumped chains, which is what section §8.1's examples exhibit; the gap between the two is exactly the redundancy Proposition 4.4 constructs. Both sentences should be rewritten, and the example is worth including, since it makes the invariant's meaning sharper rather than weaker. **[M]**

3.5 First-order theory (§6)

The interior statements are correct and correctly attributed. The residual at a metric projection onto a closed set lies in the regular normal cone; since a stratum is contained in the model set its tangent cone is smaller and the normal cones therefore reverse the inclusion, giving the orthogonality (9) at interior projections and leaving (8) as the correct general statement. The count of smooth critical points on the Zariski closure of a stratum is bounded by its Euclidean distance degree.

The boundary statement is wrong as written. The section says that when W^* lies on a boundary stratum, "the Moreau decomposition of $V - W^*$ against the tangent and polar cones of F_d at W^* then replaces (9)". Moreau's decomposition theorem requires a closed *convex* cone and its polar; the Bouligand tangent cone of a semialgebraic set need not be convex (the set $\{xy = 0\}$ at the origin is the standard example, and F_d is exactly the kind of set whose stratified boundary produces such points). As stated the decomposition does not exist. Two repairs are available and either is a one-line change: keep (8), the regular normal inequality, as the correct general statement and say that no orthogonal decomposition is claimed at boundary points; or replace the Bouligand cone by the Clarke tangent cone, which is closed and convex, and state the decomposition against it and its polar. The error does not touch the existence of the profile or any result that depends on it. **[M]**

One scope remark. The almost-everywhere uniqueness of the projection is correct, but the null set should be understood relative to the affine hull of the family space (the kernels satisfy $\mathbf{1}^\top V = \mathbf{1}^\top$, so $\mathcal{V}$ lies in a proper affine subspace of the ambient matrix space and has ambient measure zero). **[R]**

3.6 Category dependence (§7)

Proposition 7.1 is correct in substance. The inclusion of categories gives the inclusion of model sets. For strictness the argument is right: at $d = n$ the labelings are bijections, so every member of F_n^{hom} has $W(1 \leftarrow 0) = W(2 \leftarrow 1)$, a closed condition surviving the closure, while $F_n^{\mathrm{inhom}} = \mathrm{Flat}$ contains flat families with distinct one-step kernels; and the parallelogram identity gives the gap, the minimum over M being attained at the column-stochastic matrix $(A + B)/2$.

The displayed quantitative bound does not follow in the stated generality. It applies the parallelogram identity to $\|A - M\|_\pi^2 + \|B - M\|_\pi^2$ in a single inner-product norm, while the metric (4) attaches a different weight vector π_i to each source time, and the two summands being compared are indexed by $i = 0$ and $i = 1$. The two terms therefore need not live in the same inner-product space and the parallelogram identity does not apply to them as written. The necessary correction is straightforward: one can either assume common source weights for the two pairs or transition to a common norm, absorbing a norm-equivalence constant derived from the minimum positive weight. The strictness conclusion remains valid under either approach. **[M]**

3.7 Computations (§8)

Finding 5 (Upper bounds cannot certify the rank)**.** The methodology paragraph is scrupulous: the values are obtained by multistart quasi-Newton descent on softmax charts with Powell polishing, and are described as "certified upper bounds on h_d", with multistart dispersion and first-order

stationarity offered "as evidence of tightness". That is the correct status of the computation. The conclusions drawn from it are not.

An upper bound $h_3 \leq 0.026680$ is consistent with $h_3 = 0$. The statements that the profile "vanishes exactly at the generating dimension", that $\mathrm{rk}_{\mathrm{mem}} = 4$ and $\mathrm{rk}_{\mathrm{mem}} = 3$ are "recovered", and that "below the rank the charge is decisively positive" all require $h_{d-1} > 0$, which no upper bound establishes. First-order stationarity says the returned parameter is stationary in the chosen chart; multistart dispersion is evidence against easily reachable alternative minima. Neither excludes another basin, a boundary sequence, or a different labeling chart with a smaller objective, and F_d is nonconvex and semialgebraic, so the concern is not hypothetical. The manuscript's own boundary observation (projections reaching block masses of 10^{-13}) is precisely a case of the objective being approached along a direction ordinary descent does not certify.

Two repairs, of very different cost. i) For the zeros, an exact witness is free and should replace the optimiser entirely. When the family is the compression of a known lumped chain, that chain *is* a lift of the claimed dimension, so $h_d = 0$ is proved by exhibiting parameters, not by minimising to 10^{-7}; and by Proposition 4.4 the same witness pads to every higher dimension, which also removes the monotonicity anomaly of Finding 6 at a stroke. ii) For positive values, a certificate is required, which constitutes the primary analytical effort. Where it applies, the obstruction of Finding 3 supplies one directly. Otherwise a lower bound must come from global optimisation with interval arithmetic, branch-and-bound, or a sum-of-squares or polynomial-optimisation certificate of non-membership; the smallest nontrivial case, $n = 2$ and $d = 3$, would be the place to demonstrate that such a certificate is obtainable at all. Until one is, the profile's positive entries are measurements and the rank claims are conjectures supported by them. **[M]**

Finding 6 (The reported profile violates the paper's own monotonicity, and the violation is systematic)**.** Table 1 reports, for the lumped four-to-two family, $h_4 = 0.000000$ and $h_5 \leq 0.000222$. Proposition 4.4 proves $F_4 \subseteq F_5$ and hence $h_5 \leq h_4$; with h_4 at zero, h_5 must be zero as well. The printed profile is not monotone, and the manuscript passes over the fact without comment.

I could built a lumped four-to-two family and computed the profile in the time-homogeneous category with the same kind of optimiser: $h_2 = 0.054967$, $h_3 = 0.024044$, $h_4 = 0.000001$, $h_5 = 0.000099$. The non-monotonicity reproduces, on a different family, with a different implementation, at the same place and the same order of magnitude. It seems to be a property of the problem, not of one run: above the memory rank the model set admits larger and increasingly degenerate parameterisations of the same compression, the landscape flattens, and zero becomes harder to certify precisely where the theory guarantees it holds.

Taken with Finding 5 the picture is symmetric and unflattering: below the rank the optimiser returns values it cannot certify are positive, and above the rank it returns values that cannot be correct. Two repairs should accompany any reported profile, and both are free: enforce monotonicity by replacing $\hat{h}_d$ with $\min_{d' \leq d} \hat{h}_{d'}$, which the theory licenses; and report the certification tolerance alongside each value, so that "0.000000" and "≤ 0.000222" are legible as measurements rather than as numbers. **[M]**

Finding 7 (Section 8 is not reproducible, and the paper states that it is)**.** The families of §8 are specified as "a flat 2-state chain observed in full", "the compression of a randomly generated 4-state homogeneous chain through a two-and-two lumping", "the compression of a 3-state chain through a two-and-one lumping", and "a family assembled from ten independent uniformly random stochastic matrices". No matrices are printed, no random seed is given, and for the lumped families no generating distribution is stated. The grid, the weights, the optimiser and the tolerances are all given; the data are not.

The Data and code availability statement asserts the contrary: "every family, metric, and projection is specified in the text" and "the families are generated as stated". The metric, projection and tolerances are indeed specified. The families are not, and no care with the remaining ingredients recovers a number from an unspecified random draw. Availability of code on request is not a substitute: a permanent public archive containing the exact matrices, the seeds and generating procedure, the labelings and charts considered, the solver and package versions, the stopping tolerances, all starts with their incumbent values and stationarity diagnostics, and the raw outputs would make the reported upper bounds reproducible (though, by Finding 5, reproducing an upper bound still would not convert it into a proof).

I record precisely what this cost and what it did not. Every *structural* claim of §8 is testable on families of one's own, and I tested them all: the profile vanishes at the realizing dimension and is positive below it; the coherence identity and the covariance lemma hold to machine precision; the gauge path collapses the defect while remaining admissible; the non-realizable family has a positive floor, certified by the consistency defect I computed; the category inequality holds; and the projections of realizable families are interior while those of non-realizable ones are not. Not one of the reported *numbers* can be checked. Since the substantive standard applied to empirical work is reproducibility, and since the paper's own statement claims the section meets it, the discrepancy is material even though the underlying science is sound. The repair is three lines of print. **[M]**

On the category exchange rate (§8.3). The reported movement (h_3 for the lumped four-to-two family falling from 0.026680 to 0.000752 on passing from the homogeneous to the inhomogeneous category) is the paper's most interesting numerical observation, since it says one degree of time-inhomogeneity substitutes for almost all of one hidden state. It is also the observation most damaged by the two preceding findings: it is a difference of two uncertified upper bounds on one unspecified family. It deserves to be made on a specified family, with witnesses where the values are near zero, and, if the effect is generic, on a small ensemble with a reported spread.

On the word "plateau" (§8.2). Theorem 5.6 guarantees a positive floor, not flatness, and the printed values fall from 1.319266 to 0.910255, a decrease of thirty-one per cent over three dimensions; my own non-realizable family falls by twenty-nine per cent over the same range. The floor is the content; the word overstates it. **[R]**

4 Originality: what is new and what is standard

Standard, and correctly attributed. Semialgebraic geometry and Whitney stratification; the Łojasiewicz inequality; normal cones and the variational characterisation of metric projections; the Euclidean distance degree; the Ho–Kalman construction and Hankel rank; Kemeny–Snell lumpability; extension complexity and the dependence of lift size on the lifting cone. Each is cited at the point of use. The history-state Markovization of Theorem 5.7 is likewise a classical construction, correctly executed.

The paper's own. Three items survive the audit as genuinely the paper's. The gauge trivialization theorem, with its explicit one-parameter collapse onto the reset locus: I found no close antecedent for the admissible stochastic re-dressing construction, and the conclusion drawn from it is correctly derived from what is proved. The realizability gap through a Lipschitz obstruction vanishing on the model sets (the idea of the argument, which survives and indeed strengthens under the replacement proposed in Finding 3). And the category-dependence proposition with an

explicit lower bound on the gap. The coherence identity is elementary and its novelty is modest, though its use as an implementation check is a good idea.

What the memory rank measures, and what it therefore is not. The model sets are defined by the two-time block conditionals of (6) alone. The memory rank is therefore the least hidden dimension reproducing the *two-time* conditionals, which is strictly less data than the path law on which classical hidden-Markov realization theory operates; consequently the memory rank is bounded above by the minimal order of a realizing hidden Markov model, generally strictly. Combined with Finding 2, what $h_d = 0$ certifies is weaker still: approximability, at dimension d, of the two-time conditionals. The paper positions the memory rank as "the metric, temporal member" of the realization-theory family without recording either difference. Both matter for Open Problem (iii), which proposes to sharpen the dimension bound toward positive-realization bounds defined for a larger quantity on more data. **[R]**

Level. Moderate as an integrated synthesis; low to moderate for the individual results. Most proofs are a paragraph from standard facts; the value is in the architecture and in the two negative theorems that fence the construction.

5 Priority in the literature

Finding 8 (The shape of the construction is established; four positioning items are missing)**.** The manuscript places itself in metric algebraic geometry and in realization theory, and cites foundational work in both. The text omits four specific items required for proper positioning.

Approximation by models of assigned hidden dimension. Finesso, Grassi and Spreij, "Approximation of stationary processes by hidden Markov models", *Math. Control Signals Systems* **22**(1) (2010), 1–22, treat exactly the problem of minimising a discrepancy between given process data and the set of hidden Markov models of *assigned* complexity, the number of hidden states fixed in advance and the resulting minimum studied as a function of it. That is the shape of the liftability profile. The discrepancy differs (a Kullback–Leibler divergence rate there, a weighted Euclidean distance on two-time kernels here) and so does the data, but the construction is the same one, and the paper appears in the volume immediately preceding the survey the manuscript does cite from the same journal.

Distance to the stochastic part of a model variety. Casanellas, Fernández-Sánchez and Garrote-López, "Distance to the stochastic part of phylogenetic varieties" (arXiv:1912.02138; *J. Symbolic Comput.* **104** (2021)), compute Euclidean distances from data to an algebraic model variety and, separately, to the subset of it carrying a genuine probabilistic interpretation, using the semialgebraic conditions that the stochastic restriction imposes. That is the algebraic-statistics analogue of h_d (distance to the stochastic part of a model set, with the semialgebraic description doing the work) and it is the nearest existing work in the programme the manuscript places itself in.

The linear category is solved. For weighted finite automata over a one-letter alphabet, the problem of finding the closest automaton with a prescribed smaller number of states is solved exactly in the spectral norm through Adamyan–Arov–Krein theory for Hankel operators (arXiv:2102.06860, ICALP 2021). This is the linear analogue of h_d, and it is optimal rather than heuristic. The contrast sharpens the manuscript's own thesis, and it bears directly on Finding 5: it is positivity that turns the profile from a computation into an uncertified multistart optimisation, which is the temporal form of the extension-complexity phenomenon the paper already invokes.

The static analogue of the model-set geometry. Kubjas, Robeva and Sturmfels, "Fixed points of the EM algorithm and nonnegative rank boundaries", *Ann. Statist.* **43**(1) (2015), study the boundary of the set of matrices of nonnegative rank at most three as a semialgebraic object. That is the one-time analogue of the geometry of F_d, and its treatment of the boundary is directly relevant to Finding 2.

None of these anticipates the memory rank as defined here: an invariant of a two-time family, indexed by hidden dimension, with a flatness-distance at the bottom and a computable realizability obstruction. The construction stands. What is missing is the acknowledgement that its shape is not new, and a statement of what the temporal and metric refinement adds. **[R]**

6 Specific criticisms

Items are marked **[M]** where they bear on the correctness or the substance of a claim, and **[R]** where they concern presentation, attribution or scope.

[M] A vanishing profile certifies approximate liftability, not liftability
(Finding 2). The model sets are closures; a witness with $h_2 = 0$ and no lift in Θ_2^+ is recorded in the verification section, and the projections of non-realizable families approach the boundary in practice.

[M] The numerical section cannot certify the ranks it reports
(Finding 5). Upper bounds are consistent with $h_{d-1} = 0$. Supply exact witnesses for the zeros — free, since the generating chain is one — and lower-bound certificates for the positive values used to infer a rank.

[M] The reported profile violates the paper's own monotonicity proposition
(Finding 6), and the violation reproduces on an independent family with an independent implementation.

[M] The claim that the profile recovers the hidden dimension of a lumped chain is false
(Finding 4), and is refuted by the splitting construction used to prove the paper's own Proposition 4.4.

[M] Section 8.2 invokes Theorem 5.6 without verifying its hypothesis
(Finding 3). The consistency defect is necessary, not sufficient; a family with $\kappa_c = 0$ and no process is recorded, and a strictly stronger linear-programming obstruction carries the same argument.

[M] Section 8 is not reproducible, and the data statement asserts that it is
(Finding 7). The generating matrices or a seed must be printed, and the raw record archived.

[M] The Moreau decomposition at a boundary stratum is invalid,
because the Bouligand tangent cone of a semialgebraic set need not be convex. Retain (8), or pass to the Clarke tangent cone and its polar.

[M] The quantitative bound of Proposition 7.1 does not follow
when the metric attaches different weight vectors π_0 and π_1 to the two pairs compared. The strictness conclusion survives.

[R] Four positioning items are missing
(Finding 8): approximation by hidden Markov models of assigned dimension; distance to the stochastic part of a model variety; the exact solution of the linear analogue by Adamyan–Arov–Krein theory; and the semialgebraic geometry of nonnegative-rank boundaries.

[R] The invariant-theoretic language of Remark 3.5 exceeds what is proved,
which is Corollary 3.4.

[R] The re-dressing group is the similarity group of quasi-realizations
(Finding 1) and should be named as such.

[R] The higher-coherence tower is asserted without a complex or a proof.

[R] Lipschitz stability of h_d is not consistency of the memory rank,
which is read off at a discontinuity and needs a margin and a threshold sequence.

[R] The memory rank is an invariant of the two-time family only,
and, by Finding 2, of approximability rather than membership; Open Problem (iii) compares quantities defined from different data.

[R] Example 5.2 omits the positivity and closure step
required by the definition of Θ_d^+ — the same step that produces the witness of Finding 2.

[R] The closure clause of Theorem 5.7 should use the universal bound $D_{\max}$,
not the family-specific D, which a boundary family does not possess.

[R] The almost-everywhere uniqueness statement should be relative to the affine hull
of the family space.

[R] "Plateau" overstates Theorem 5.6,
which gives a positive floor; the reported values fall by thirty-one per cent across the dimensions displayed.

[R] The arithmetic floor of the gauge path is not recorded;
below $\varepsilon \approx 3 \times 10^{-3}$ on a five-point grid the computed defect rises again.

[R] The proof of Proposition 4.5 contains a garbled clause
("1-Lipschitz squared-root").

7 Questions to the author

1. Your model sets are closures. Prove that $V \in F_d$ implies an exact d-state lift in Θ_d^+, or rename the invariant. Consider the flat family generated by $M = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: it has $h_2 = 0$ and no lift in Θ_2^+. Is the gap confined to families with degenerate one-time marginals?

2. Characterise a boundary sequence whose block masses tend to zero. Which conditional columns remain identifiable in the limit, and what statistical meaning attaches to $h_d = 0$ attained only through such a sequence?

3. Why does an optimiser value $h_3 \leq 0.026680$ not prove $h_3 > 0$? What form of certificate would prove non-membership in F_3, and can you produce one in the smallest nontrivial case $n = 2$, $d = 3$?

4. For every claimed zero in Table 1, exhibit the parameters. For a lumped chain the generating chain is such a witness, so the zeros need no optimisation at all — and, by Proposition 4.4, the witness pads to every higher dimension, which disposes of the h_5 entry.

5. Table 1 reports $h_4 = 0.000000$ and $h_5 \leq 0.000222$, which Proposition 4.4 forbids. What certifies the $d = 4$ entry if the $d = 5$ entry is wrong, and what tolerance should accompany each cell?

6. Apply your own Proposition 4.4 to split a hidden state of a three-state chain and lump the two copies together. The observed family is unchanged and its memory rank is three while the generating dimension is four. Restate the claim of the abstract and of §8.1 accordingly.

7. Does $\kappa_c(V) = 0$ imply that V is process-induced? I record a family with $\kappa_c = 0$ exactly whose pairwise-compatibility linear program is infeasible. What are the missing conditions, and would the linear-programming defect $\delta(V)$ — which vanishes on every exact compression and is Lipschitz in V by your own argument — not give a strictly stronger version of Theorem 5.6?

8. For the family of §8.2, what is κ_c? Theorem 5.6 is invoked for it without its hypothesis being checked.

9. Print the generating matrices and initial laws of the three lumped families of §8, or a seed and the exact generating procedure, and archive the raw optimiser record.

10. State the assumptions under which Moreau's decomposition applies. Is the Bouligand tangent cone of a semialgebraic model set necessarily convex, and what is the correct boundary statement?

11. Repair the bound of Proposition 7.1 when π_0 and π_1 define different weighted norms.

12. What exactly does Corollary 3.4 prove, and what further structure would be needed before speaking of instability, a categorical quotient, or a ring of invariants?

13. How does the memory rank relate to the minimal order of a hidden Markov model realizing the full path law, to nonnegative rank, and to Hankel rank? Which implications hold and which fail?

14. How does the liftability profile relate to the approximation of a process by hidden Markov models of assigned complexity, and to the computation of distances to the stochastic part of a model variety? What does your formulation add?

15. In the linear category the analogous problem is solved exactly by Adamyan–Arov–Krein theory. What is the obstruction in the stochastic category, and is it the same positivity obstruction that extension complexity describes?

16. Formulate the coherence analogy as an actual complex. Does δ_V square to zero on higher cochains?

17. What statistical assumptions would make the integer memory rank consistently estimable from noisy estimates of V, as distinct from estimating h_d?

18. Can the memory rank decrease under refinement of the time grid? Is there any monotonicity theorem in the grid?

8 Technical conclusion

The theorem presented in this manuscript are correct as a mathematical statement, with two local exceptions that are repairable in a line each: the Moreau decomposition invoked at a boundary stratum, where the relevant tangent cone need not be convex, and the quantitative bound of Proposition 7.1, which applies a parallelogram identity across two different weighted norms.

What the theorems are said to mean is, in four places, stronger than what they say. The model sets are closures, so a vanishing profile certifies approximability at a hidden dimension and not the existence of a lift; I record a family with $h_2 = 0$ that no admissible lift realizes, and the projections of non-realizable families approach the boundary in practice, so the distinction is operational. The numerical section produces upper bounds, which cannot establish the strict positivity on which every rank claim rests; and independently, above the rank the same computation returns values the theory forbids, so the rank is uncertified on both sides. The claim that the profile recovers the generating dimension of a lumped chain is false, and is refuted by the paper's own splitting construction; the memory rank is the minimal realizing dimension, which is the correct and sharper invariant. And the realizability obstruction is one-sided, while section §8.2 uses it as though it were two-sided: a family with consistent marginals and no process is recorded here, together with a strictly stronger obstruction, computable by linear programming, that vanishes on every exact compression and carries the same Lipschitz argument to the same conclusion.

To these must be added a computational section that the text cannot reproduce although its own data statement says it can, and a group of positioning items: the shape of the construction (a distance from data to the set of objects explainable at an assigned hidden dimension, indexed by that dimension) is established in both the realization and the algebraic-statistics literature, and the linear analogue is solved exactly, which sharpens rather than diminishes the paper's thesis about positivity.